

COURSE CODE: IT411

AVIATION CAPSTONE PROJECT 2

Mission Plan & Project Roadmap

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PRE-REQ: IIT327

LECTURE: 3 UNITS

DURATION: 54 HOURS (PROJECT-BASED / RESEARCH)

MISSION PARAMETERS

The culminating implementation phase of Capstone Project 2. Focuses on system completion, validation, oral defense, and deployment of an approved IT-based aviation system.



30%

Gauge chart



25%

Gauge chart



25%

Gauge chart



20%

Gauge chart

Flight Rules: Code of Conduct

Operational Directives



Attendance: Maximum 20% absence threshold before automatic failure. 15-minute tardiness grace period.

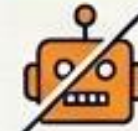


Accountability: Mandatory group roles (Project Leader, Dev, Researcher). Individual accountability applies.



Consultations: Verified consultation logs and revision matrices are required evidence of progress.

Ethical Directives



AI Disclosure: AI allowed for grammar/debugging ONLY. Strict ban on fabricating data or manuscript content.

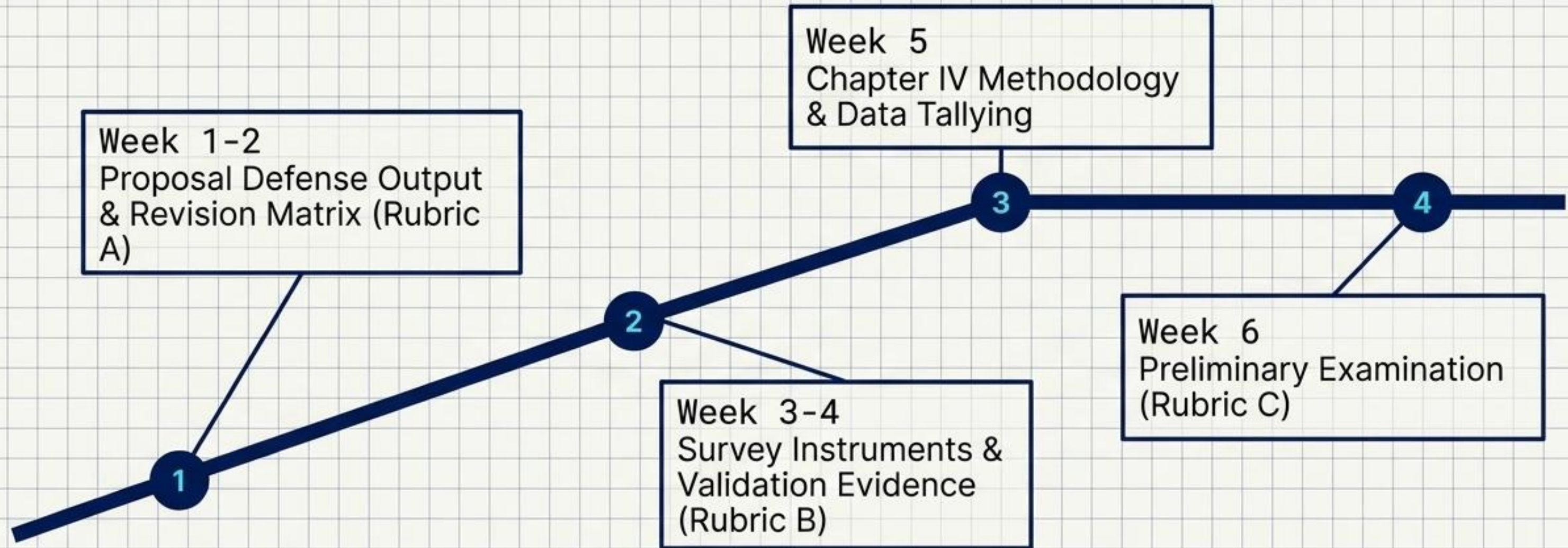


Academic Integrity: Zero tolerance for plagiarized code, falsified validation forms, or fake survey responses.



Data & IP: Mandatory informed consent, responsible handling of personal data, and respect for institutional IP.

Phase 1: Instrumentation & Calibration (Weeks 1-6)



PHASE OBJECTIVE: Establish baseline readiness, validate evaluation instruments, and lock in preliminary Chapter IV data with at least 80% compliance.

The Triangulated Evaluation Matrix

Pre-Assessment 	ISO/IEC 25010 	TAM (Technology Acceptance Model) 
Target Respondents Target Beneficiaries	Target Respondents IT Experts	Target Respondents End-Users
Core Purpose Establish the baseline problem and context before system implementation.	Core Purpose Evaluate software engineering quality and technical robustness.	Core Purpose Measure behavioral intention and usability.
Key Metrics Current inefficiencies, needs analysis.	Key Metrics System Quality Requirements and Evaluation (SQuaRE) characteristics.	Key Metrics Perceived Usefulness, Perceived Ease of Use.

Mission Checkpoints: Phase 1

Checkpoint A (Proposal & System Demo)

TARGET 4.0



Complete approved manuscript and panel comments presented.



Clear, logical alignment among objectives, features, and documentation.



Accurate revision matrix with a feasible plan.



Functional prototype clearly demonstrating relevant objectives.

Checkpoint B (Instruments & Validation)

TARGET 4.0



Survey items strongly aligned with ISO 25010, TAM, and objectives.



Appropriate validators and fully documented forms.

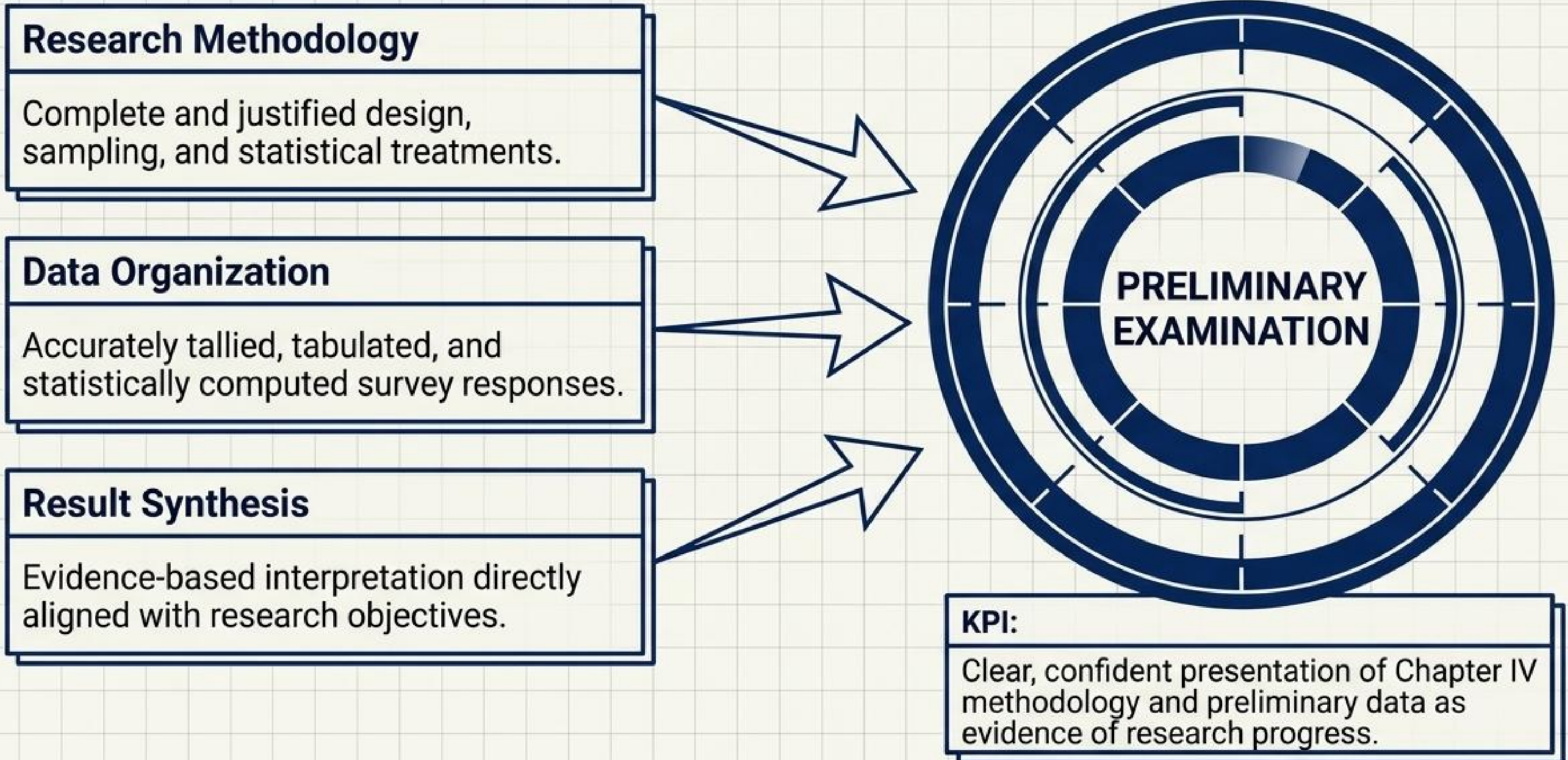


Correctly computed validity and reliability test results.

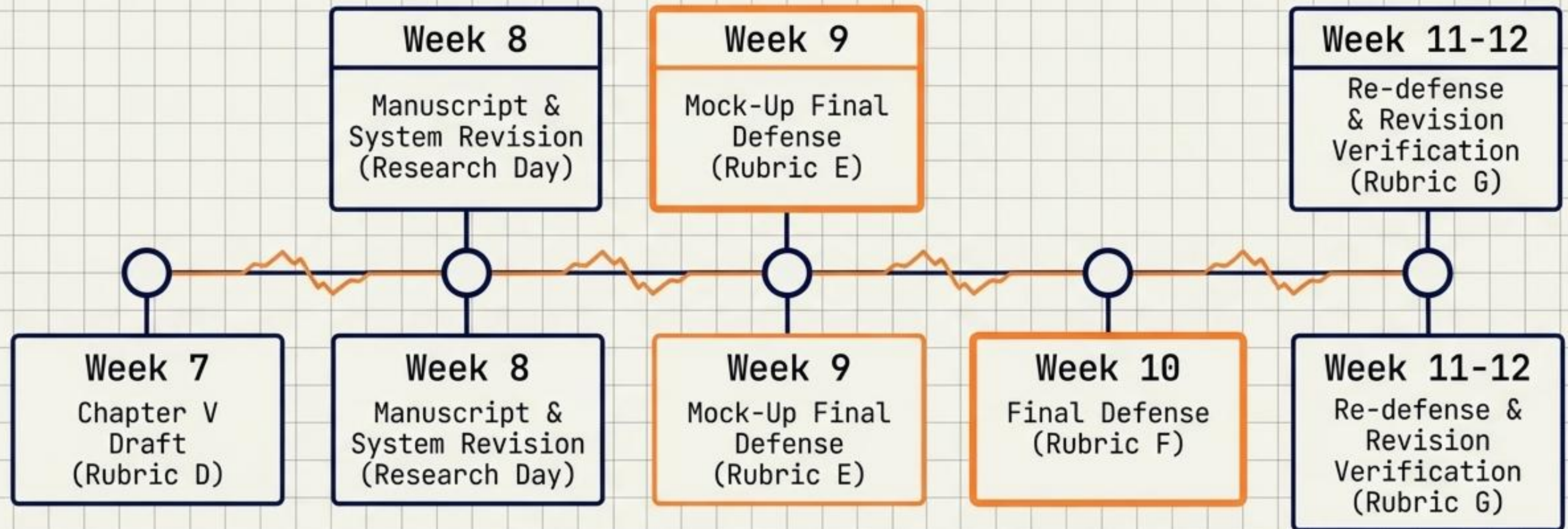


Ethical safeguards (consent, confidentiality) strictly in place.

Checkpoint C: Chapter IV & Preliminary Data

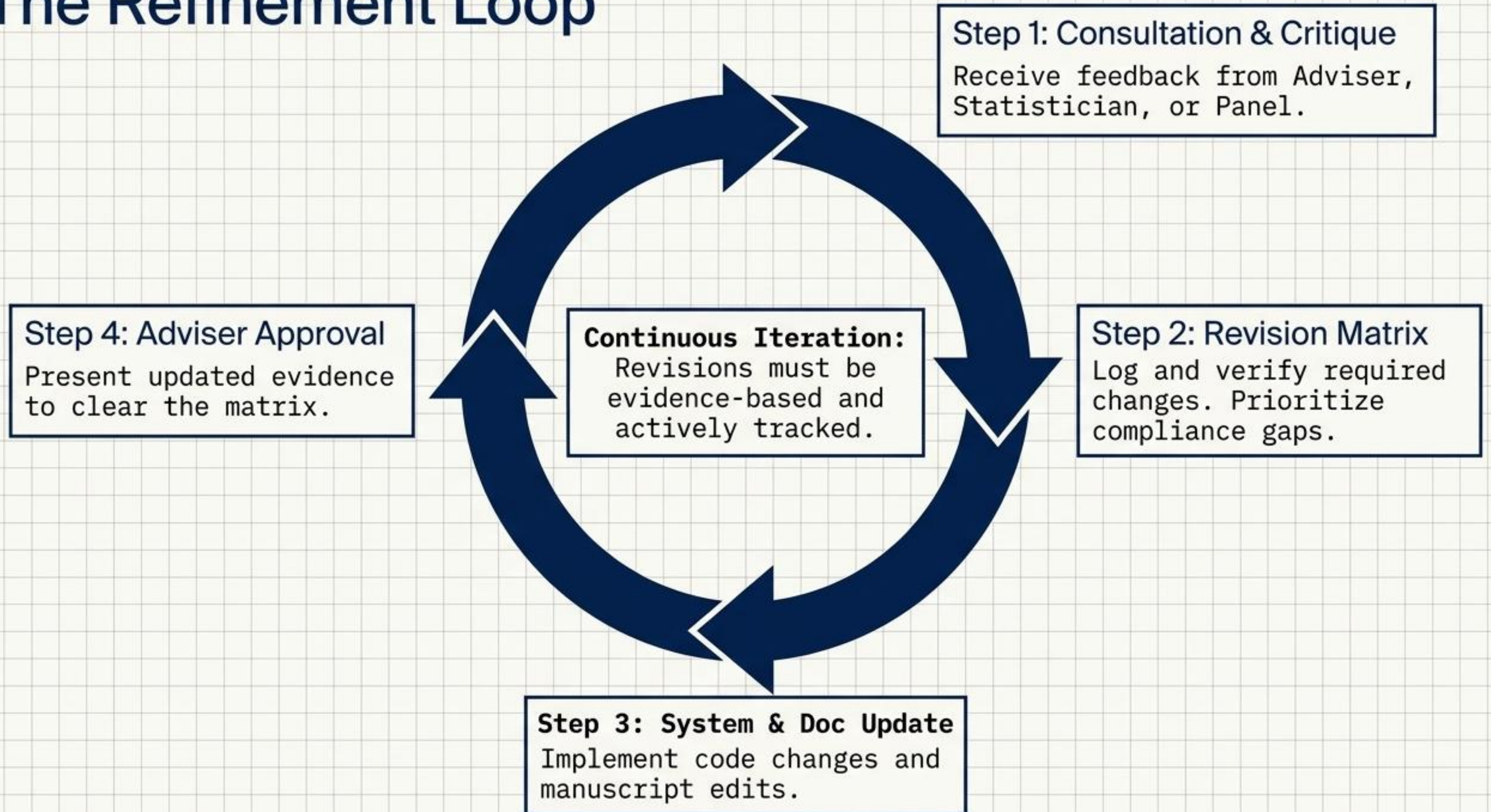


Phase 2: The Proving Grounds (Weeks 7-12)



PHASE OBJECTIVE: Synthesize final conclusions, endure rigorous panel evaluation, and demonstrate a fully functional IT capstone system.

The Refinement Loop



The Defense Gauntlet

Preliminary Exam (Week 6)	Mock Defense (Week 9)	Final Defense (Week 10)
Audience Faculty & Peers.	Audience Simulated Panel (Peers/Adviser).	Audience Official Capstone Panel.
Primary Focus Chapter IV. Data gathering, statistics, and initial results.	Primary Focus Presentation flow, system stability, rubrics feedback.	Primary Focus Complete system validation (80%+ functional) and Chapter I-V defense.
Required Output Data tables, statistical outputs, methodology draft.	Required Output Full slide deck, working prototype, peer-critique forms.	Required Output Completed manuscript, live system demonstration, evidence-based Q&A.

Mission Checkpoints: Phase 2

Checkpoint D (Chapter V & Revisions)

TARGET 4.0



Logical, evidence-based conclusions rooted in Chapter IV findings.



Consultation logs and revision matrices complete and properly signed.



Manuscript is coherent, formatted to NAAP/ICS layout, and defense-ready.

Checkpoint E & F (Mock & Final Defense)

TARGET 4.0



System is stable, functional, and aligns with 80%+ of stakeholder requirements.

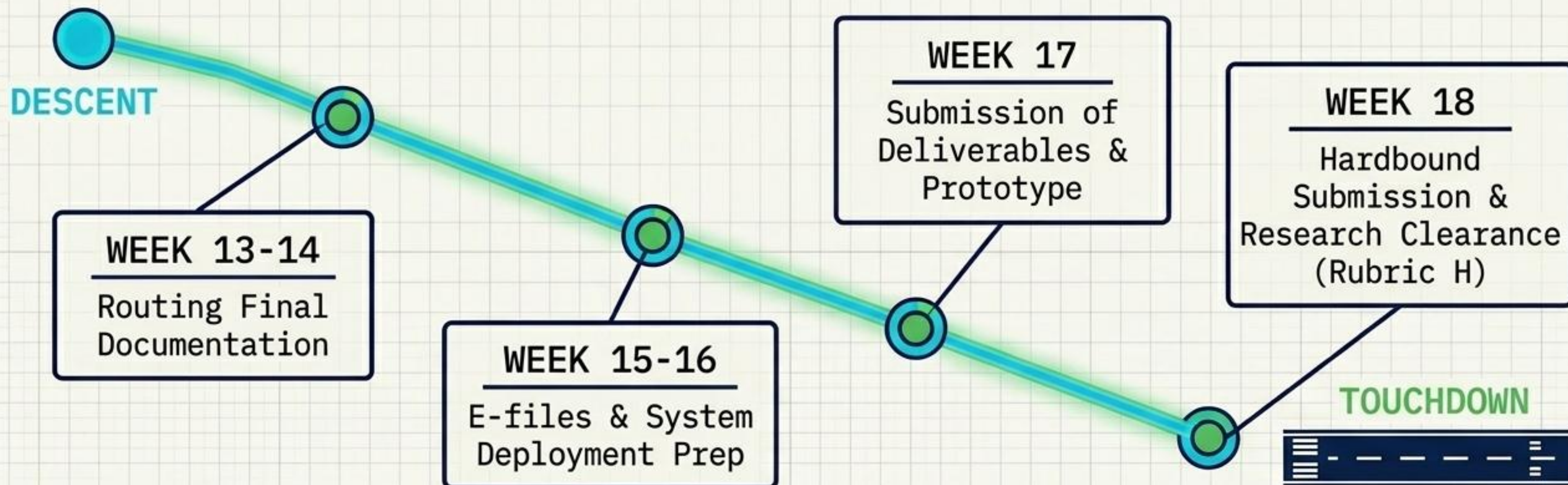


Accurate, evidence-based responses to all panel questions.



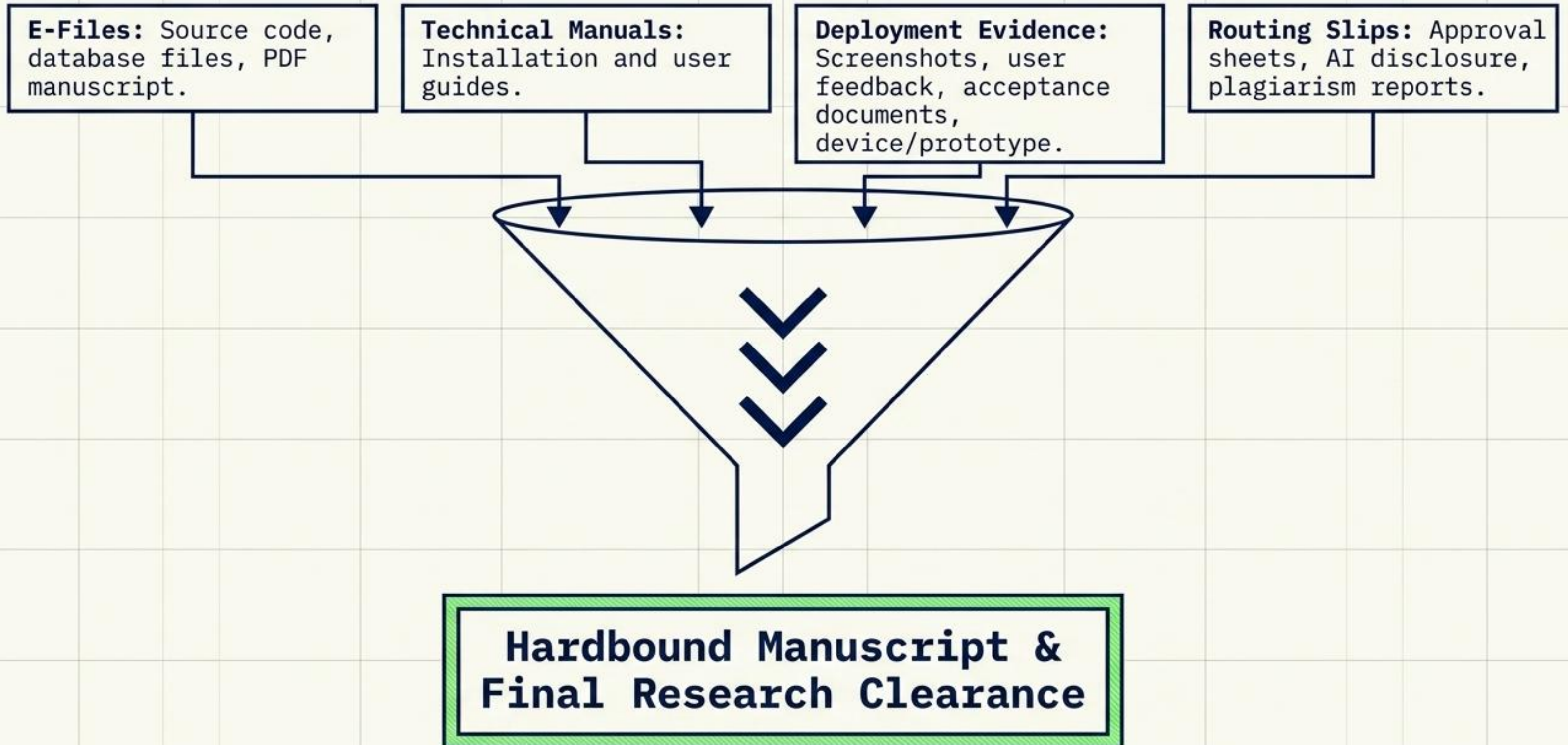
Professional, time-conscious oral presentation with seamless system demo.

PHASE 3: DEPLOYMENT & HANGAR (WEEKS 13-18)



PHASE OBJECTIVE: Secure authorized signatures, deploy the live system, archive source code, and achieve institutional clearance.

The Archiving Funnel

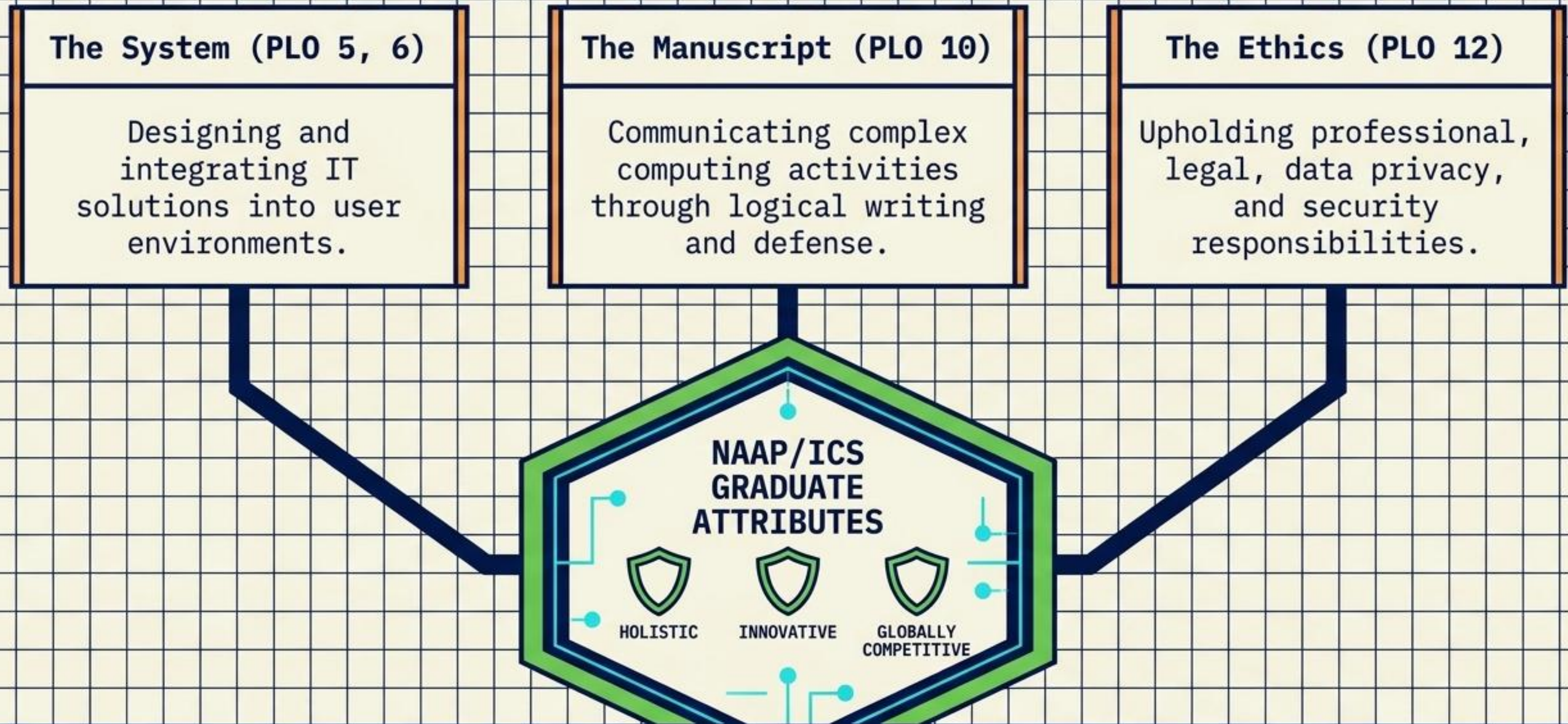


Mission Checkpoints: Final Clearances

Checkpoint G (Re-defense / Revisions) TARGET 4.0	
✓	All panel comments completely addressed with clear screenshot/document evidence.
✓	System revisions actively improve functionality based on feedback.
✓	Output is strictly ready for final routing and deployment.

Checkpoint H (Deployment & Clearance) TARGET 4.0	
✓	Routing forms, approval sheets, and signatures fully complete.
✓	Organized digital folders containing manuals, source code, and credentials.
✓	Complete deployment reports, user acceptance, and prototype evidence.
✓	Hardbound manuscript and research clearance successfully filed.

Payload Delivery: The Ultimate Output



You haven't just built an application; you have proven your readiness for the aviation IT industry.